

Aakash Ramakirushnan Selvam

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SUMMARY

Data Analyst with 2.5 years of experience in DevOps, Full-Stack Development, and Data Management. Currently pursuing an MS in Business Analytics and AI at UT Dallas (May 2025). Experienced in developing applications, analyzing data, and providing insights to improve efficiency and decision-making. Interested in applying AI, predictive analytics, and visualization to solve business problems.

TECHNICAL SKILLS

MLOps & Model Deployment: CI/CD for ML, Model Packaging (Docker), Model Monitoring | **Programming & Web Development:** Python, R, SQL, Node.js, React, Express.js, HTML, CSS, C | **Cloud & DevOps:** AWS, Kubernetes, Docker, Ansible, Jenkins, CI/CD | **Data Science & Machine Learning:** Apache Spark, Hadoop, XGBoost, Neural Networks | **Data Analysis & Visualization:** Tableau, Excel, MongoDB, MySQL, PostgreSQL, Git.

PROFESSIONAL EXPERIENCE

Tata Consultancy Services, Chennai, India | Client: Siemens, US | Role: Full Stack Developer August 2020 – December 2022

- Developed and deployed a human well-being application using **React.js**, **Node.js**, collaborating with clients to frame requirements and sell the solution, successfully scaling it to **20,000+ active users** and driving a **30% increase in engagement**.
- Analyzed user behavior data using **MongoDB and BigQuery**, making data-driven changes to the application based on client feedback and usage patterns, leading to continuous improvements in performance and user satisfaction.

Tata Consultancy Services, Chennai, India | Client: Internal Project | Role: DevOps Engineer

- Automated infrastructure with Ansible**, reducing deployment time by **30%**, and using **Jenkins for CI/CD**, increasing deployment by **20%**.
- Performed log and data trend analysis** using **SQL**, identifying critical system issues and reducing downtime by **25%**, while leveraging **Alops** tools like Prometheus for proactive anomaly detection and automated incident management

Tata Consultancy Services, Chennai, India | Client: Internal Project | Role: Intern

January 2020 – March 2020

- Optimized resource utilization and reduced downtime by **50%** by **automating pod scaling in Kubernetes**, ensuring efficient workload management and seamless application performance even during traffic spikes.
- Enhanced deployment efficiency by **30%** by leading the implementation of **Docker** containerization, streamlining **CI/CD workflows**, and enabling faster, more reliable application delivery across multiple environments, reducing manual intervention and deployment errors.

ACADEMIC PROJECT EXPERIENCE

Site selection and Visualization, Kimberly-Clark

March 2025

- Built an interactive **Tableau** dashboard integrating census, economic, and risk assessment data for real-time site selection.
- Ranked among the Top 5 teams** by developing a **data-driven site selection model** that reduced decision-making time by **70%**, leveraging geospatial analysis, network optimization, and multi-source data integration to identify optimal manufacturing locations.

Optimization of Bike-Sharing Membership Conversion

January 2024 – May 2024

- Analyzed 1.4 GB** of Divvy bike-sharing data using **big data** tools to identify trends and optimize membership conversion strategies.
- Implemented **dynamic pricing and targeted** promotions, using data insights to drive customer engagement and increase membership.

Credit Risk Assessment and Default Prediction

January 2024 – May 2024

- Developed an **XGBoost model** for **credit default prediction**, achieving an **8.01%** default rate while maximizing revenue.
- Conducted data modeling, statistical analysis, hyperparameter tuning, and SHAP analysis to evaluate model performance, comparing XGBoost, neural networks, and ensemble models. The XGBoost model outperformed **neural networks** by **11.6%** in prediction accuracy.

Customer Purchase Behavior Analysis in Retail Book Stores

August 2023 – December 2023

- Examined customer purchase behaviors at two retail bookstores using **Poisson Regression** and **NBD Regression**, optimizing model selection with **log-likelihood**, **AIC**, and **BIC** to achieve **94% predictive accuracy**.
- Leveraged customer analytics to design and implement data-driven marketing strategies, increasing purchase frequency by **35%**.

LEADERSHIP EXPERIENCE

President, Vaalmeengal Tamil Sangam – The University of Texas at Dallas

October 2024 – Present

- Directed the end-to-end planning and execution of **all Indian cultural events** at UT Dallas, demonstrating strong leadership in managing a **25-member team** to ensure seamless collaboration and successful event delivery.
- Oversaw and managed a **\$20,000 fund**, including financial support from the **university and sponsors**, ensuring the efficient allocation of resources and the successful execution of community-building initiatives.

Alumni Core Committee, Petit Seminaire Higher Secondary School, India

January 2023 - Present

- Spearheaded the transformation of the 176-year legacy** of my high school as a key member of the **Alumni Core Committee**, playing a pivotal role in organizing the first alumni meet in 175 years and uniting a **global alumni network** to create a thriving, connected, and impactful community for generations to come.

EDUCATION

The University of Texas at Dallas, Richardson, Texas - *Master of Science, Business Analytics*

August 2023 – May 2025

Sri Manakula Vinayagar Engineering College, Puducherry, India – *B.Tech, Electronics and Communication Engineering*

September 2020

AWARDS AND ACHIEVEMENTS

- Represented India in Japan during a student exchange program – JENESYS 2.0, sponsored by the Japanese government.
- National medalist in Bharatanatyam (Indian classical dance) on roller skates.
- Runner-up in Smart India Hackathon 2019.